

IoT Prototype — Arduino Project

Temperature & Humidity Monitoring System

CodeAlpha IoT Internship — Task 3
Submitted by: Seif Eldaby
Platform: Arduino Uno + DHT11 Sensor

1. Project Overview

This project demonstrates a functional IoT prototype for real-time temperature and humidity monitoring using an Arduino Uno microcontroller and a DHT11 digital sensor. The system continuously reads environmental data and displays it on a 16x2 LCD screen. It also implements a visual and audio alert system: a green LED indicates normal conditions, a red LED activates for high temperature alerts, and a buzzer sounds when both temperature and humidity exceed critical thresholds — simulating a real-world IoT environmental monitoring node for applications such as server rooms, greenhouses, warehouses, and home automation systems.

2. Project Objectives

- Read temperature (°C) and relative humidity (%) in real time using a DHT11 sensor.
- Display live sensor data on a 16x2 LCD with I2C interface.
- Implement a threshold-based alert system using LEDs and a passive buzzer.
- Output all readings to the Serial Monitor for debugging and logging.
- Demonstrate the core sensing and actuation layer of a complete IoT system.

3. Components & Hardware

Component	Qty	Specification / Role
Arduino Uno R3	1	ATmega328P MCU — main controller
DHT11 Sensor	1	Digital temp (0–50°C ±2°C) & humidity (20–90% RH)
16x2 LCD (I2C)	1	Displays temperature and humidity readings
Green LED	1	Normal condition indicator
Red LED	1	High temperature alert indicator
Passive Buzzer	1	Audio alert for critical conditions

10kΩ Resistor	1	DHT11 pull-up on data line
220Ω Resistor	2	Current limiting for LEDs
Breadboard	1	Prototyping platform
USB Cable (Type-B)	1	Power supply and code upload
Jumper Wires	Several	Circuit interconnections

4. Circuit Diagram & Wiring

DHT11 Sensor Connections:

Pin 1 (VCC) → Arduino 5V. Pin 2 (Data) → Arduino Digital Pin 2, with a 10kΩ pull-up resistor between Data and VCC. Pin 4 (GND) → Arduino GND. The pull-up resistor is essential for reliable single-wire communication with the DHT11.

LCD I2C Module Connections:

VCC → Arduino 5V. GND → Arduino GND. SDA → Arduino A4 (SDA). SCL → Arduino A5 (SCL). The I2C interface reduces wiring to just 2 data lines, making the circuit significantly cleaner than a parallel LCD connection.

LED and Buzzer Connections:

Green LED: Anode → 220Ω → Pin 8; Cathode → GND. Red LED: Anode → 220Ω → Pin 9; Cathode → GND. Passive Buzzer: Positive → Pin 10; Negative → GND.

Component Pin	Arduino Pin	Notes
DHT11 VCC	5V	Power supply
DHT11 Data	Digital 2	+ 10kΩ pull-up to 5V
DHT11 GND	GND	Ground reference
LCD VCC	5V	Power supply
LCD GND	GND	Ground reference
LCD SDA	A4	I2C data line
LCD SCL	A5	I2C clock line
Green LED (+)	Pin 8	Via 220Ω resistor
Red LED (+)	Pin 9	Via 220Ω resistor
Buzzer (+)	Pin 10	Direct connection
All LED/Buzzer (-)	GND	Common ground

5. Arduino Source Code

The following code requires two libraries: DHT sensor library by Adafruit (install via Arduino Library Manager: Sketch → Include Library → Manage Libraries → search 'DHT sensor library') and LiquidCrystal_I2C library.

[illegible]

[illegible]

```

        // NORMAL CONDITIONS
        digitalWrite(GREEN_LED, HIGH);
        digitalWrite(RED_LED, LOW);
        noTone(BUZZER_PIN);
        Serial.println(">> Status: All parameters normal.");
    }

    delay(2000); // Read every 2 seconds (DHT11 minimum interval)
}

```

6. Code Explanation

6.1 Library Imports

The DHT.h library handles the single-wire communication protocol required by the DHT11 sensor. Wire.h and LiquidCrystal_I2C.h handle I2C communication with the LCD module. These are standard Arduino libraries available through the Library Manager.

6.2 Sensor Reading and Validation

dht.readHumidity() and dht.readTemperature() return float values. The isnan() check validates each reading — DHT11 communication can occasionally fail due to timing issues, and processing NaN (Not a Number) values would produce incorrect behavior. If a read fails, the system prints an error, displays it on the LCD, and retries after 2 seconds.

6.3 LCD Display Logic

The LCD is cleared each iteration and updated with fresh readings. Line 1 shows temperature (e.g. 'Temp: 27.3 C') and Line 2 shows humidity (e.g. 'Hum: 65.0 %'). The lcd.print() function supports floats with precision control — the '1' argument limits output to one decimal place for clean formatting on the 16-character display.

6.4 Two-Level Alert System

The alert system uses two thresholds. When temperature alone exceeds 35°C, the red LED activates. When both temperature exceeds 35°C AND humidity exceeds 80%, the buzzer also activates with a 1kHz tone. This two-level design simulates a real IoT alert hierarchy — a visual warning first, escalating to audio for critical combined conditions. The delay(2000) respects the DHT11's minimum sampling interval of 1–2 seconds.

7. Expected Output

Condition	Green LED	Red LED	Buzzer	LCD Line 2 Status
Temp < 35°C, Hum < 80%	ON	OFF	OFF	All Normal

Temp >= 35°C, Hum < 80%	OFF	ON	OFF	High Temp Alert
Temp >= 35°C, Hum >= 80%	OFF	ON	ON (1kHz)	CRITICAL Alert

Serial Monitor Output Sample:

```
=====
IoT Temp & Humidity Monitor
CodeAlpha Internship | Task 3
=====
Temperature: 27.3 C | Humidity: 62.0 %
>> Status: All parameters normal.
Temperature: 36.1 C | Humidity: 75.0 %
>> ALERT: High Temperature!
Temperature: 37.4 C | Humidity: 83.5 %
>> ALERT: High Temperature!
>> CRITICAL: High Temp + High Humidity!
```

8. IoT Extension Possibilities

This prototype forms the sensing and actuation core of a complete IoT system. The following extensions would complete it into a fully connected IoT node:

Extension	Hardware/Technology	Benefit
Wi-Fi Connectivity	ESP8266 / ESP32	Send data to cloud (ThingSpeak, Firebase)
Mobile Alerts	IFTTT / Telegram Bot API	Push notifications to smartphone
Data Logging	SD Card Module	Local storage of historical readings
Web Dashboard	Node-RED / Grafana	Real-time visualization & analytics
Battery Power	LiPo + TP4056 module	Wireless deployment anywhere

9. Conclusion

This prototype successfully demonstrates the fundamental principles of an IoT environmental monitoring system. The DHT11 sensor provides reliable temperature and humidity data, the Arduino processes readings and applies intelligent threshold logic, and the output devices (LCD, LEDs, buzzer) provide clear multi-modal feedback. The code is modular, well-commented, and handles sensor errors gracefully — reflecting professional embedded systems practices. With the addition of wireless connectivity, this prototype would become a fully deployable IoT monitoring node suitable for smart homes, greenhouses, server rooms, or industrial environments.

References

- [1] Aosong Electronics. (2023). DHT11 Temperature & Humidity Sensor Datasheet. Aosong (Guangzhou) Electronics.
- [2] Adafruit Industries. (2023). DHT Sensor Library for Arduino.
github.com/adafruit/DHT-sensor-library
- [3] Arduino Documentation. (2024). tone() Function Reference.
arduino.cc/reference/en/language/functions/advanced-io/tone/
- [4] Blum, J. (2013). Exploring Arduino: Tools and Techniques for Engineering Wizardry. Wiley.
- [5] Schwartz, M. (2016). Internet of Things with Arduino Cookbook. Packt Publishing.